SOFTWARE REQUIREMENTS SPECIFICATION

AI-Powered Academic Advisor

Project Title	AI-Powered Academic Advisor
Document Type	Software Requirements Specification (SRS)
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Prepared By	Divyanshi Gupta Shreya Singh Tanmay Shekhar Harshit Singh Bharti
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1. Introduction

1.1 Purpose

This SRS defines the functional, non-functional, interface, data, and modeling requirements for an AI-Powered Academic Advisor. The system is intended to help students make informed academic decisions by analyzing their academic performance, subjects, interests, skills, goals, and preferences.

1.2 Scope

The system collects a student profile and academic information, analyzes performance and preferences through an AI recommendation component, identifies academic strengths and weak areas, recommends suitable subjects or learning actions, generates personalized study guidance, and tracks academic progress.

1.3 Objectives

• Provide personalized academic recommendations.
• Analyze academic performance and identify weak areas.
• Recommend suitable subjects, electives, and learning priorities.
• Generate personalized study plans and learning resources.
• Help students monitor progress toward academic goals.
• Explain the factors behind important recommendations.

2. Overall Description

2.1 Intended Users

User	Main Responsibilities
Student / Learner	Creates profile, enters academic information, completes assessments, views recommendations, follows study guidance, and tracks progress.
Faculty / Academic Advisor	Reviews student information and recommendations to support mentoring and academic planning.
Administrator	Manages users, subjects, academic rules, learning resources, and system data.

2.2 Product Perspective

The system is a personalized academic decision-support platform. Student information is converted into structured academic data, evaluated against subject, skill, and learning knowledge, and returned as academic insights, recommendations, priority areas, and progress guidance.

2.3 High-Level System Flow

Student Profile + Academic Data → **AI Analysis Engine** → **Performance Analysis** → **Weak-Area Detection** → **Academic Recommendations** → **Study Plan / Resources** → **Progress Tracking**

2.4 Major Product Functions

Authentication, profile management, academic data management, assessment, AI performance analysis, subject/elective recommendation, weak-area identification, study-plan generation, learning-resource recommendation, progress tracking, dashboard/reporting, and administrator data management.

3. Functional Requirements

ID	Requirement	Description
FR1	Registration & Login	The system shall allow students and authorized users to register and securely authenticate.
FR2	Profile Management	The system shall allow students to enter and update education, semester, interests, skills, goals, and preferences.
FR3	Academic Data Management	The system shall collect marks, grades, attendance, completed subjects, and other relevant academic information.
FR4	Academic Assessment	The system shall collect assessment responses related to learning preferences, strengths, difficulties, and academic goals.
FR5	Performance Analysis	The system shall analyze academic records and identify strengths, weak subjects, and performance trends.
FR6	Academic Recommendation	The system shall recommend suitable subjects, electives, learning priorities, or academic actions based on the student's profile and performance.
FR7	Suitability / Priority Score	The system shall generate a score, priority level, or confidence indicator for important recommendations.
FR8	Study Plan Generation	The system shall generate a personalized study plan based on academic goals, weak areas, priorities, and available time.
FR9	Learning Resources	The system shall recommend relevant courses, notes, practice material, projects, or other learning resources.
FR10	Progress Tracking	The system shall record learning progress and allow students to update completion status.
FR11	Dashboard & Reports	The system shall display academic insights, recommendations, weak areas, study plans, resources, and progress in a consolidated dashboard.
FR12	Academic Data Administration	The administrator shall be able to add and update subjects, skill mappings, academic rules, and learning resources.

4. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR1	Security	Student credentials and academic data shall be protected from unauthorized access.
NFR2	Performance	Normal academic analysis and recommendation requests should return results within an acceptable interactive response time.
NFR3	Reliability	The system shall provide consistent results for the same input and handle failures gracefully.
NFR4	Usability	The interface shall be clear, responsive, and understandable for students.
NFR5	Scalability	The system shall support growth in students, subjects, skills, resources, and academic records.
NFR6	Maintainability	Academic data, recommendation logic, and AI components shall be modular and maintainable.
NFR7	Availability	The deployed application should be available whenever students need academic guidance.
NFR8	Explainability	Important recommendations should show relevant performance, preference, or matching factors.
NFR9	Privacy	Only information necessary for academic guidance shall be collected, and access shall be role-controlled.

5. External Interface Requirements

5.1 User Interface

The web interface shall provide registration/login, profile setup, academic data entry, assessment, performance analysis, recommendations, study plans, learning resources, and progress tracking. The layout shall be responsive.

5.2 Software Interface

The frontend shall communicate with backend services through APIs. The backend shall access the database and AI recommendation/analysis component. The AI component shall accept structured student attributes and return academic insights and recommendations.

5.3 Communication Interface

Deployed communication shall use HTTPS. API requests and responses should use a structured format such as JSON.

6. System and Data Requirements

6.1 Hardware Requirements

Component	Minimum / Recommended
Processor	Intel Core i3 / equivalent or better
RAM	4 GB minimum; 8 GB recommended for local AI/ML development
Storage	At least 10 GB free space
Display	1366 × 768 or higher
Internet	Required for web deployment and external learning resources

6.2 Software Requirements

Component	Requirement
Operating System	Windows / Linux / macOS
Frontend	Web-based frontend application
Backend	Server-side application and REST API
Database	Database for students, subjects, academic records, assessments, recommendations and progress
AI/ML	Academic performance analysis and recommendation engine
Development	VS Code or equivalent; Git/GitHub
Browser	Chrome / Edge / Firefox or modern equivalent

6.3 Core Data Entities

Entity	Important Attributes
User / Student	user_id, name, email, education, semester, interests, goals, preferences
Subject	subject_id, subject_name, semester, credits, category, prerequisites
Academic Record	record_id, user_id, subject_id, marks, grade, attendance, semester
Assessment	assessment_id, user_id, questions, responses, score
Recommendation	recommendation_id, user_id, subject_id/action_id, score, reasons
Learning Resource	resource_id, title, subject_id/skill_id, type, provider, link
Progress	progress_id, user_id, subject_id/skill_id, status, completion_percentage

7. Data Flow Diagrams (DFD)

7.1 DFD Level 0 — Context Diagram

The complete application is represented as a single process. External entities provide student and academic information and receive analysis, recommendations, study guidance, reports, and progress outputs.

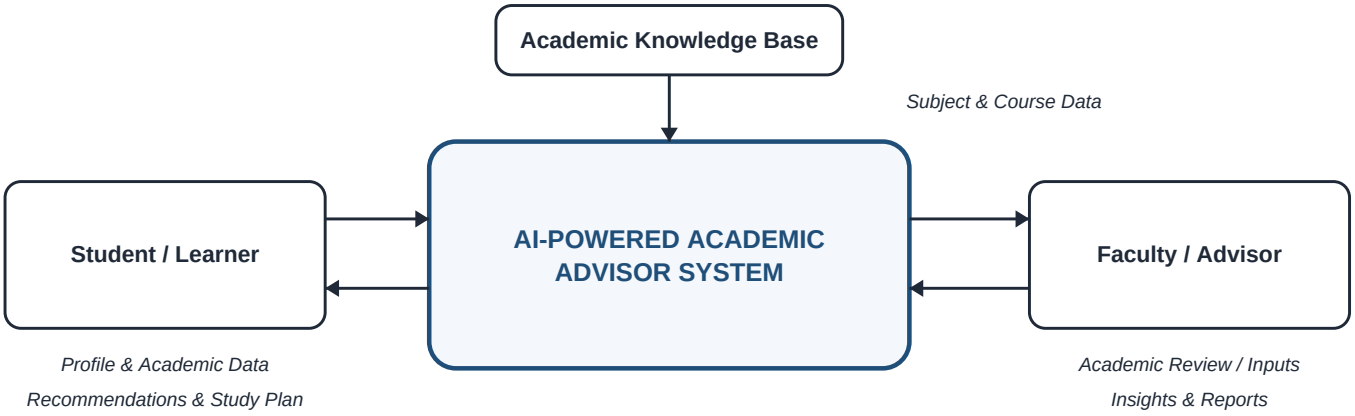
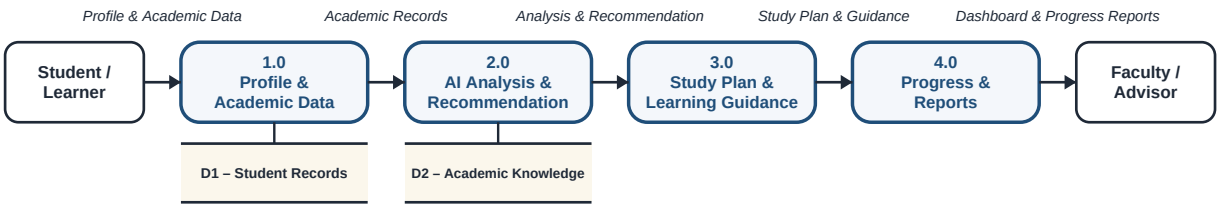


Figure 1. DFD Level 0 — Context Diagram

7.2 DFD Level 1 — Major Processes

Level 1 decomposes the system into academic profile/data management, AI analysis and recommendation, study-plan/resource generation, and progress/reporting processes. The main logical data stores are Student Academic Records and Academic Knowledge Base.



Logical flow: Student → Profile & Academic Data → AI Analysis & Recommendation → Study Plan & Learning Guidance → Progress & Reports. Faculty/Advisor may review academic insights and recommendations where authorized.

Figure 2. DFD Level 1 — Major Processes and Data Stores

8. Decision Table

The decision table defines how the system should interpret combinations of academic performance, learning priority, goal alignment, and profile completeness. It can be used as a rule layer alongside the AI recommendation component.

Rule	Performance	Priority Gap	Goal Match	Profile Complete	Action
R1	High	Low	High	Yes	Recommend advanced/enrichment activities and maintain current plan.
R2	Low / Medium	High	High	Yes	Prioritize weak subjects, recommend resources, and create a focused study plan.
R3	High	High	High	Yes	Recommend the target path but emphasize missing prerequisite skills or subjects.
R4	Low	—	Low	Yes	Do not prioritize the option; show alternatives better aligned with performance and goals.
R5	—	—	—	No	Request profile/academic-data completion before generating final recommendations.

8.1 Decision Logic

A practical implementation may combine normalized factors such as academic performance, subject/skill match, interest or goal alignment, prerequisites, attendance, and assessment fit. A generic formulation is: **Overall Recommendation Score = weighted performance fit + goal/interest match + prerequisite fit + assessment fit + learning priority**. Exact weights should be validated using project data.

8.2 Example Output

For a student whose target subject or academic path has high goal alignment but a large weak-area gap, the system should still present the option when appropriate, while emphasizing the missing prerequisites and generating a focused learning roadmap.

9. Use Case Summary

Actor	Primary Use Cases
Student / Learner	Register/Login; Manage Profile; Enter Academic Data; Take Assessment; Analyze Performance; Get Academic Recommendations; View Priority Areas; Generate Study Plan; View Resources; Track Progress.
Faculty / Academic Advisor	Login; Review Student Profile; Review Academic Insights; Review Recommendations; Interpret Weak Areas; Support Academic Planning.
Administrator	Login; Manage Users; Manage Subjects; Manage Academic Rules/Skill Mappings; Manage Learning Resources; View Reports.

10. Requirement Traceability Matrix

Requirement	Related Module / Model	Primary Actor
FR1–FR2	Authentication & Profile	Student
FR3	Academic Data Module	Student
FR4	Assessment Module	Student
FR5–FR7	AI Analysis & Recommendation Engine	Student / Faculty
FR8–FR9	Study Plan & Learning Resource Module	Student
FR10–FR11	Progress & Dashboard	Student / Faculty
FR12	Administration Module	Administrator
NFR1, NFR9	Security & Privacy	All
NFR2–NFR8	Application Quality	All

11. Constraints, Risks and Future Enhancements

11.1 Constraints

• Academic curricula, subject structures, prerequisites, and learning resources may change over time.
• Recommendation quality depends on the quality and coverage of the underlying academic data/model.
• External learning resources may change or become unavailable.
• AI recommendations are guidance and should not replace institutional academic policies or professional academic advising.

11.2 Risks and Mitigations

Risk	Impact	Mitigation
Incomplete academic data	Irrelevant or inaccurate recommendations	Validation, completeness checks, and reassessment.
Outdated subject/resource data	Incorrect study plan or recommendations	Administrator updates and periodic review.
Model bias	Unsuitable recommendations for some students	Evaluate diverse student profiles and expose matching factors.
Over-reliance on AI	Poor academic decision-making	Clearly present results as guidance and encourage verification with faculty/institutional rules.
Incorrect performance interpretation	Wrong priority areas	Use validated scoring rules, input validation, and human review where required.

11.3 Future Enhancements

• Automatic extraction of academic data from report cards or institutional records.
• Integration with real-time curriculum and course information.
• Personalized multi-semester academic roadmaps.
• Faculty/mentor matching and appointment support.
• Evidence-based skill assessment through projects, quizzes, or coding tests.
• Feedback-driven improvement of recommendations.
• Early-warning alerts for declining performance or attendance.

12. Conclusion

The AI-Powered Academic Advisor provides a structured approach to academic planning. It combines student profile, academic records, assessments, goals, and academic knowledge to produce performance insights, prioritized academic recommendations, personalized study guidance, learning resources, and progress tracking. The DFDs, decision table, use-case summary, and traceability matrix provide clear software-engineering representations of data movement, decision logic, user interactions, and requirement coverage.

End of Software Requirements Specification